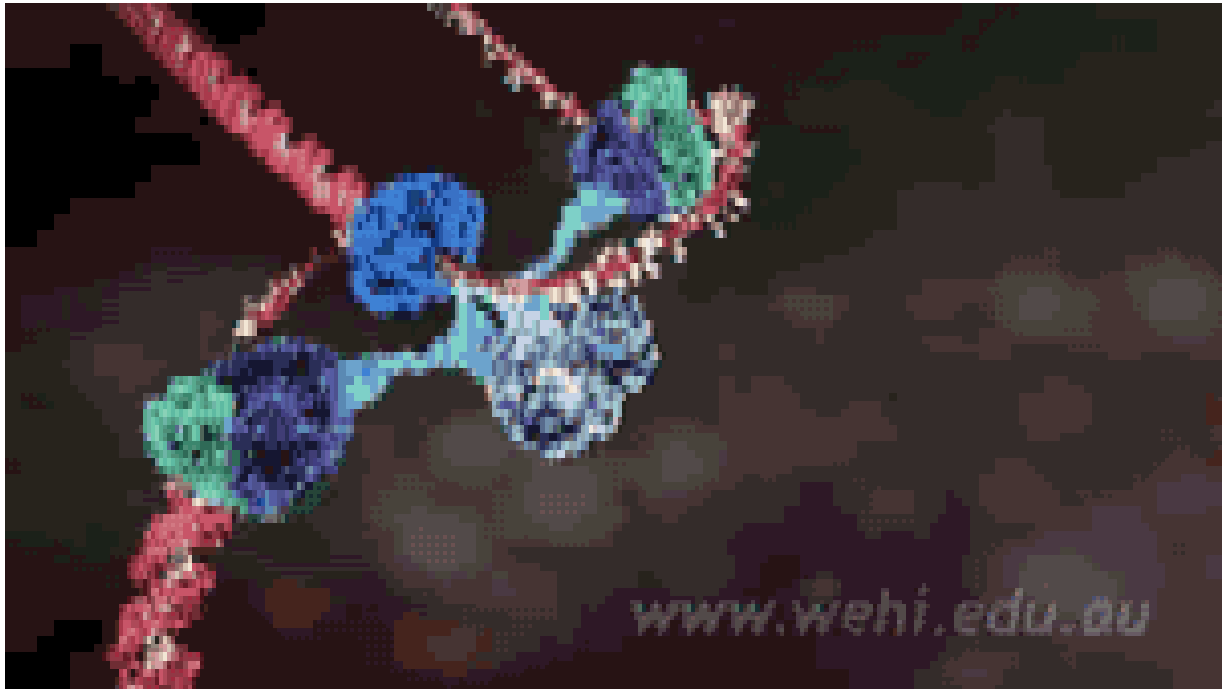
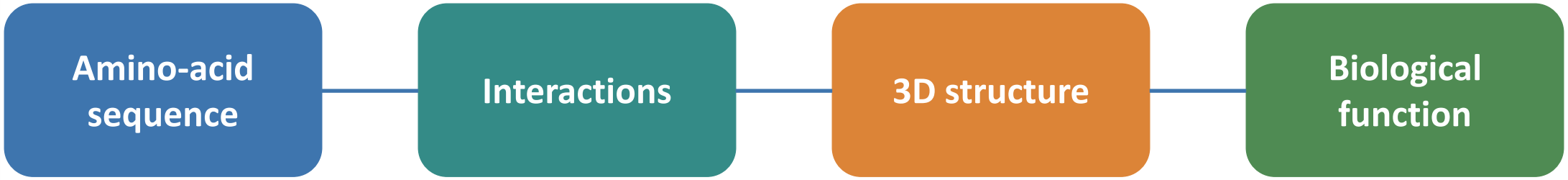
Biomolecules - Proteins

Lecture 11



The central idea: sequence → structure → function

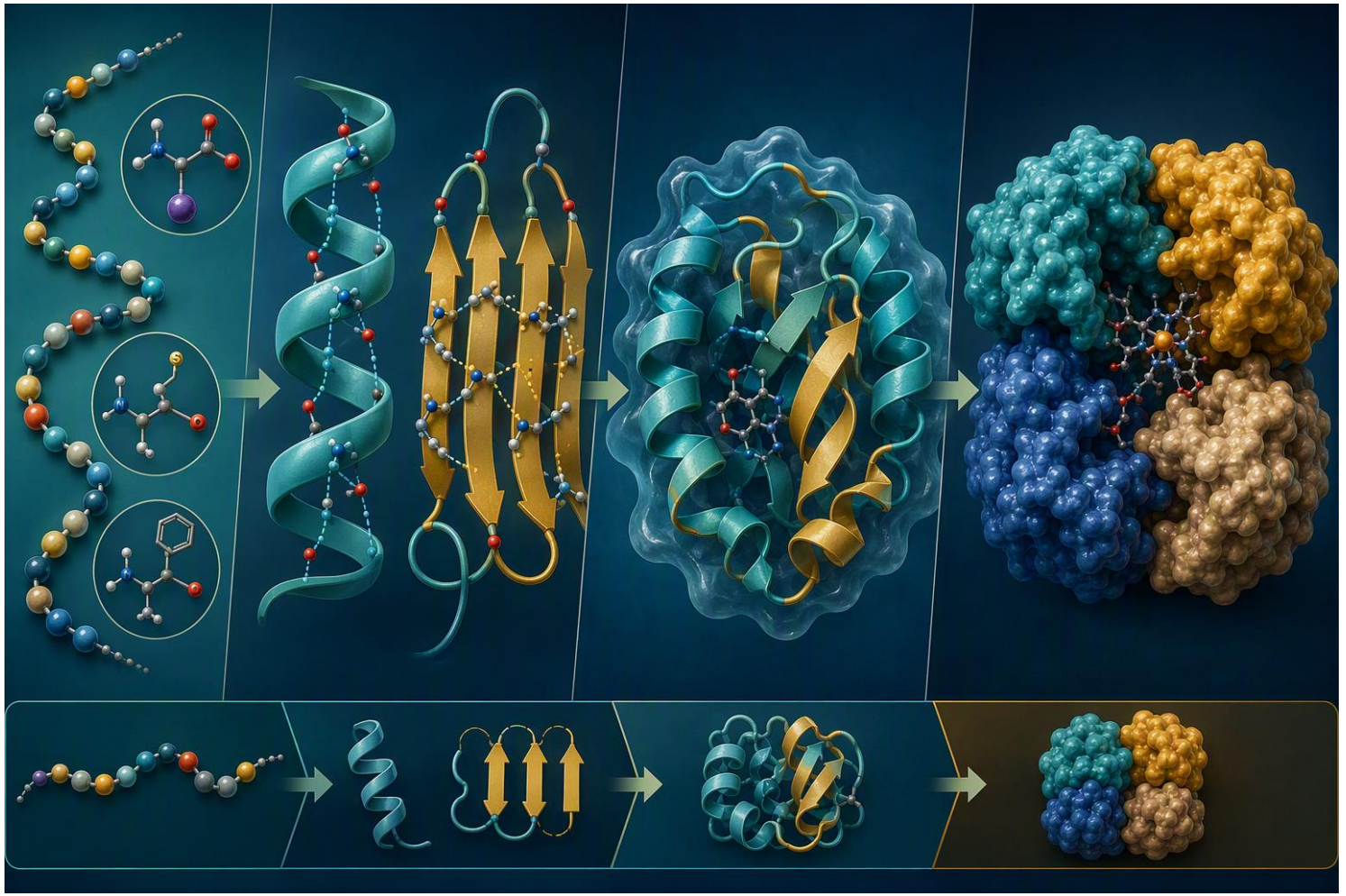


Environment & mutations can perturb every step

Hierarchy of protein structure

Primary	amino-acid sequence
Secondary	α-helices and β-sheets
Tertiary	3D fold of one polypeptide
Quaternary	assembly of multiple chains

From sequence to functional structure



Primary
amino-acid sequence and peptide bonds

Secondary
local α-helices, β-sheets and turns

Tertiary + quaternary
3D fold and association of subunits

Structure enables function!

Structural configuration of proteins

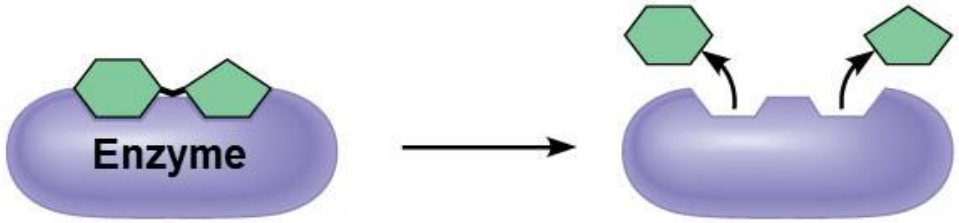
PROTEIN TYPE	PROTEIN FUNCTION
Structural proteins	Provide mechanical support for tissues e.g. collagen
Transporters	Carry molecules e.g. Hemoglobin and oxygen transport
Enzymes	Catalyze reactions
Receptor proteins	When on cell surface - communicate information from outside to inside cells
Signaling proteins	Control cellular processes
Gene regulatory proteins	Turn on and off gene expression into proteins e.g. transcription factors
Membrane transporters	Regulate movement of molecules across cellular membranes e.g. ion channels

Functions of Proteins

Enzymatic proteins

Function: Selective acceleration of chemical reactions

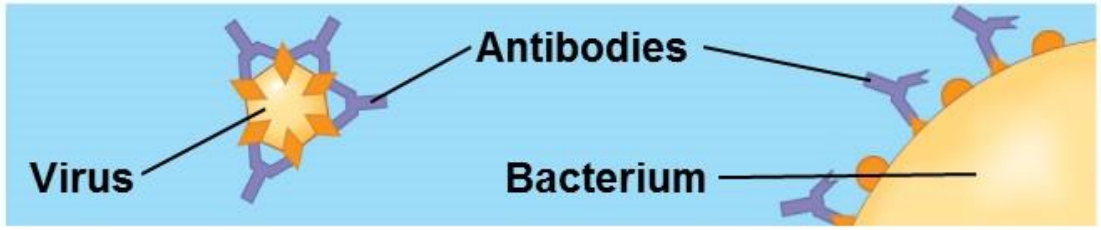
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins

Function: Protection against disease

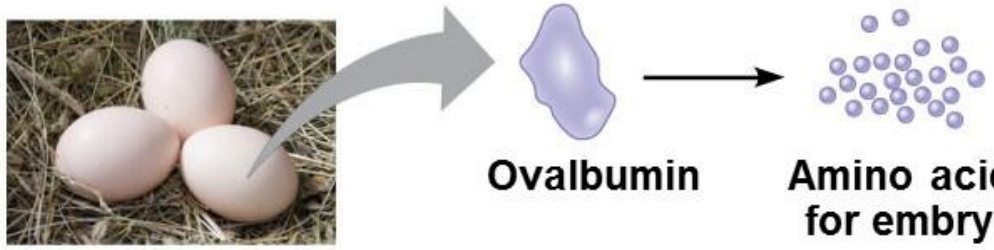
Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

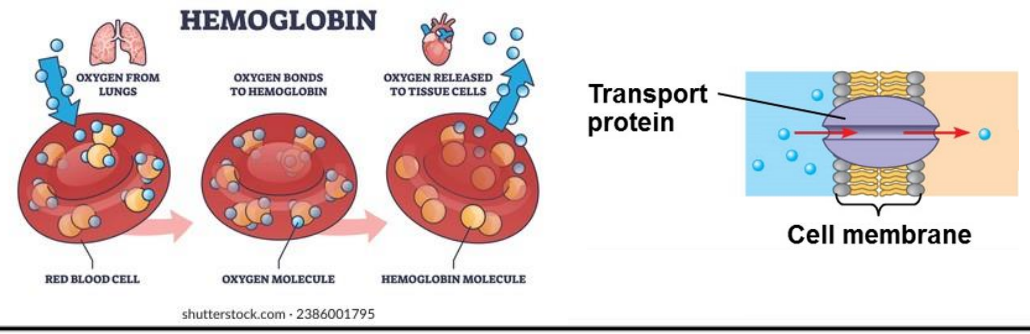
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins

Function: Transport of substances

Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across cell membranes.

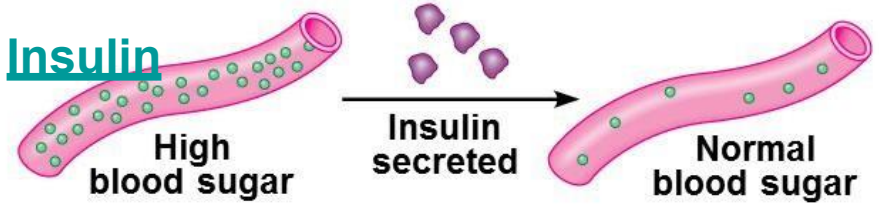


Functions of Proteins

Hormonal proteins

Function: Coordination of an organism's activities

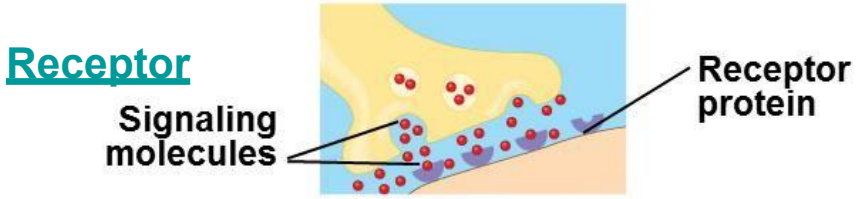
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration



Receptor proteins

Function: Response of cell to chemical stimuli

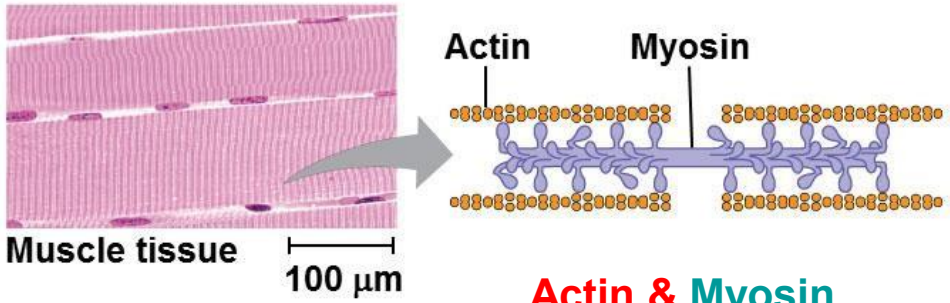
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

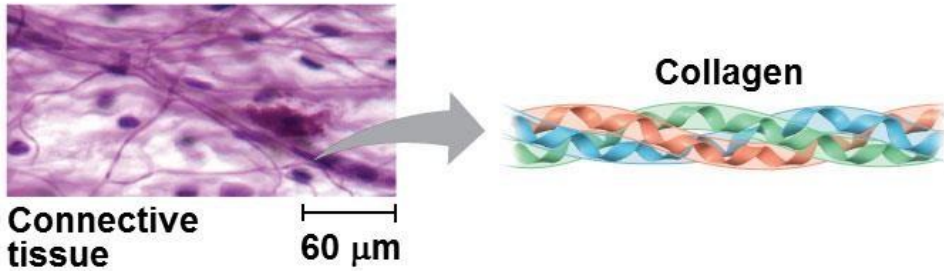
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



Structural proteins

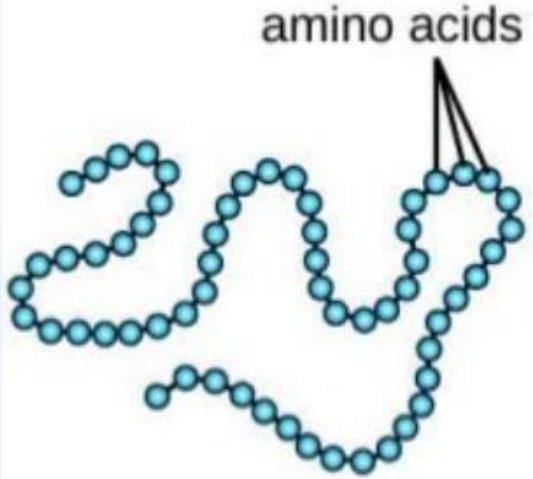
Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.





Protein Denaturation



Primary Protein Structure
Sequence of a chain of amino acids

Protein Folding

Gain of function



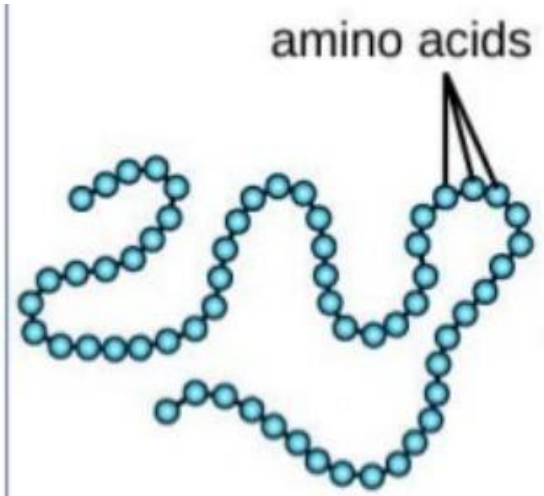
Quaternary Protein Structure
protein consisting of more than one amino acid chain



Quaternary Protein Structure
protein consisting of more than one amino acid chain

Protein unfolding

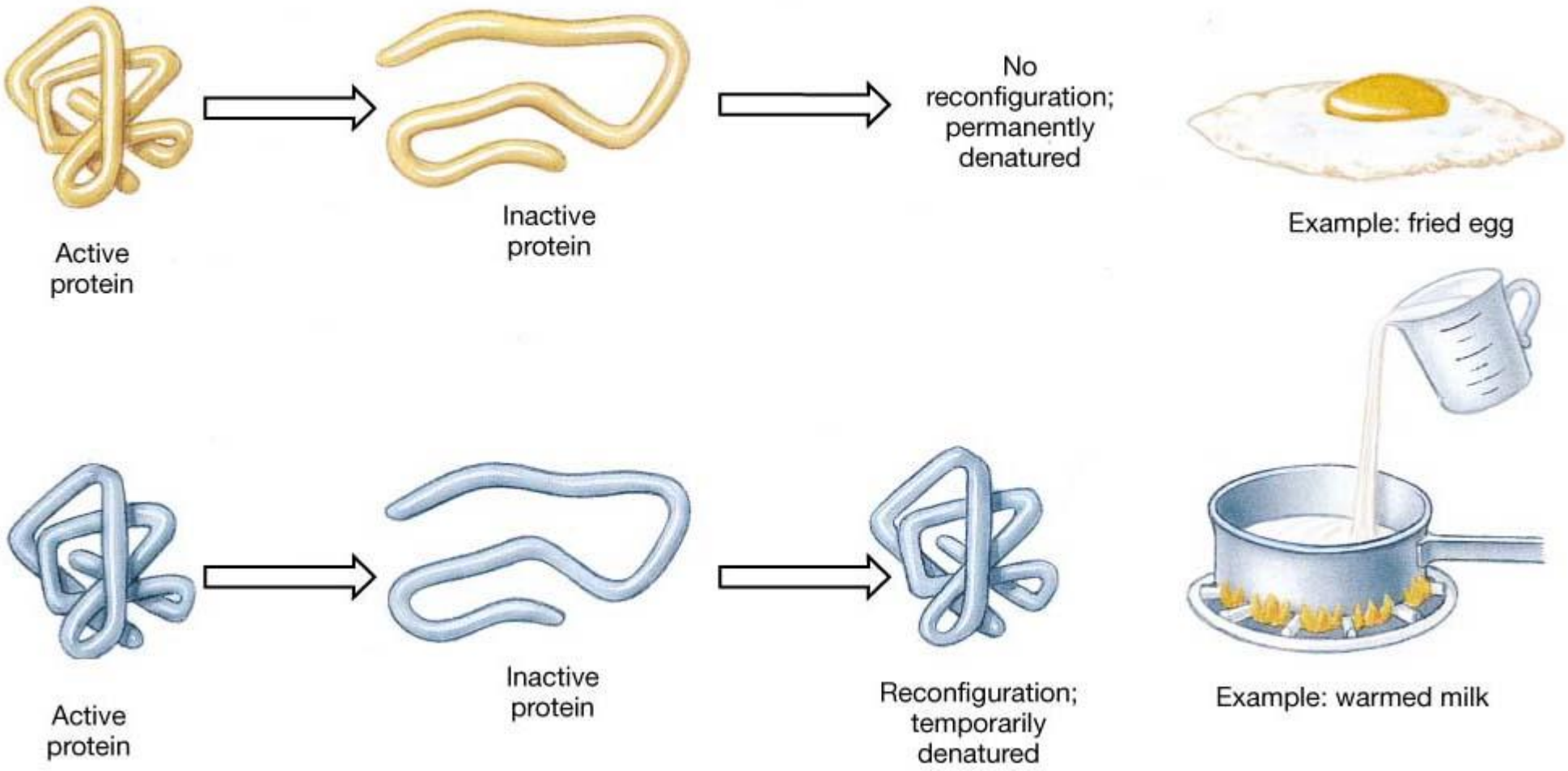
loss of function



Primary Protein Structure
Sequence of a chain of amino acids

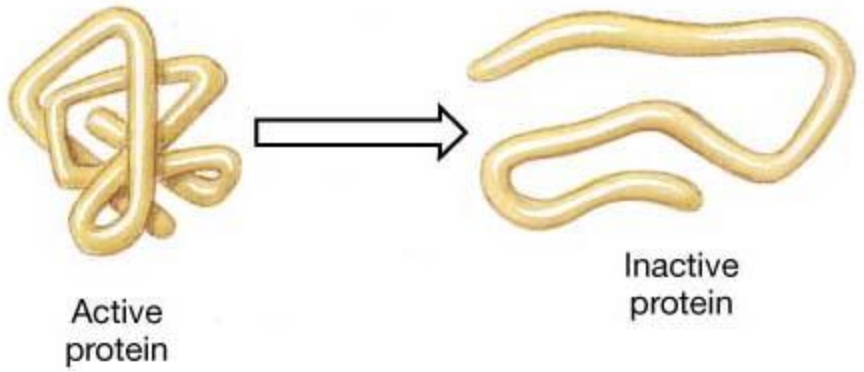
Denaturation is loss of native higher-order structure and usually biological activity.

Denaturation of proteins



Denaturation of proteins

- Heat: Breaks the non-covalent bonds and leads to the protein making aggregates with itself
- pH: Changes the protonated state, which in turn affects there binding in the protein chain.
- Salt and organic solvents: They make interaction with parts of proteins and break them apart. E.g. ethanol can break apart hydrophobic core in protein



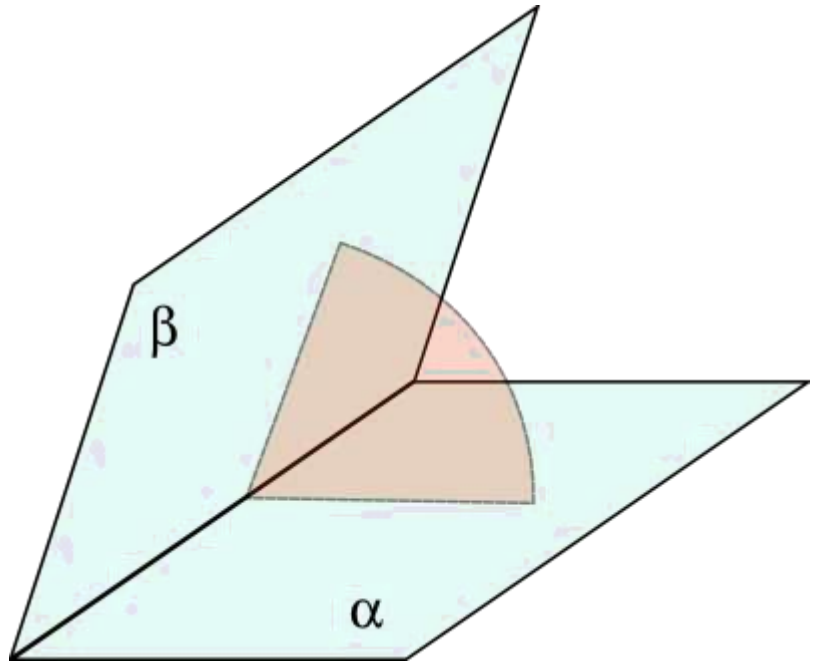
Protein denaturation can lead to lethal diseases, e.g., prion disease or Alzhimer disease, occur due to misfolded proteins.



Ramachandran Plots

Dihedral angles

- It is the angle between two intersecting planes.

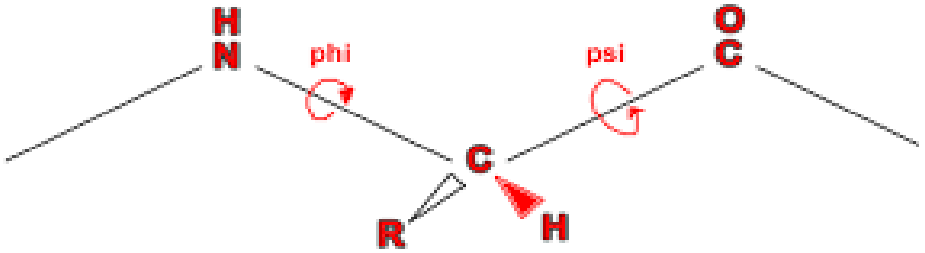


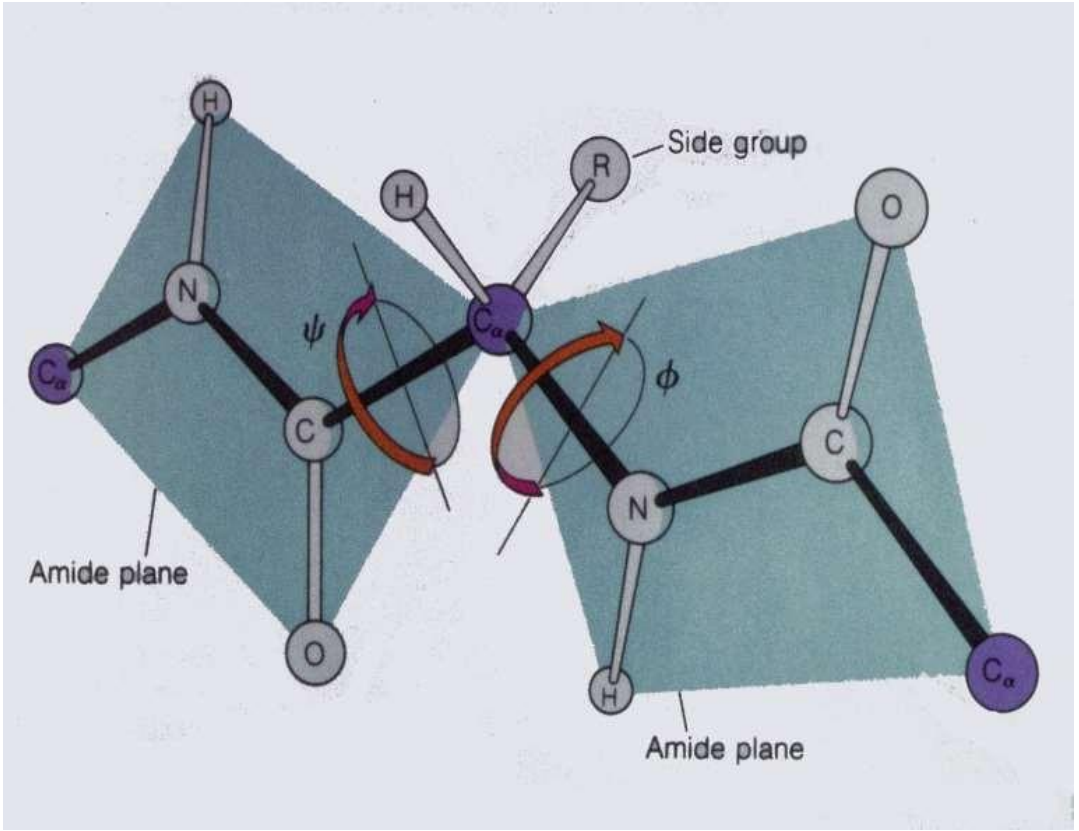
Polypeptide chain conformations

- The backbone or main chain of a protein refers to the atoms that participate in peptide bonds, ignoring the side chains of the aminoacid residues.

- The only reasonable free movements are rotations around the Cα – N bond(measured as Φ) and the Cα –C bond (measured as Ψ).

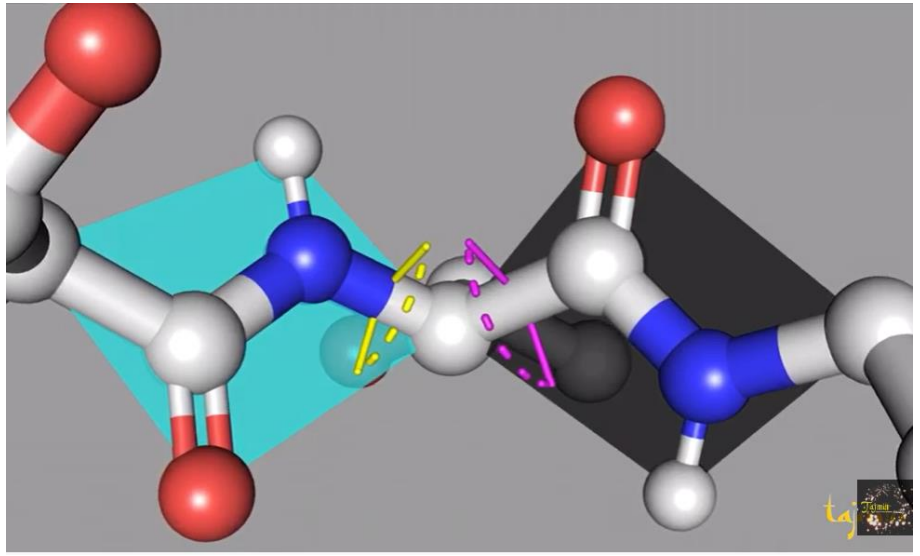
- The conformation of the backbone can therefore be described by the torsion angles (also called dihedral angles or rotational angles).

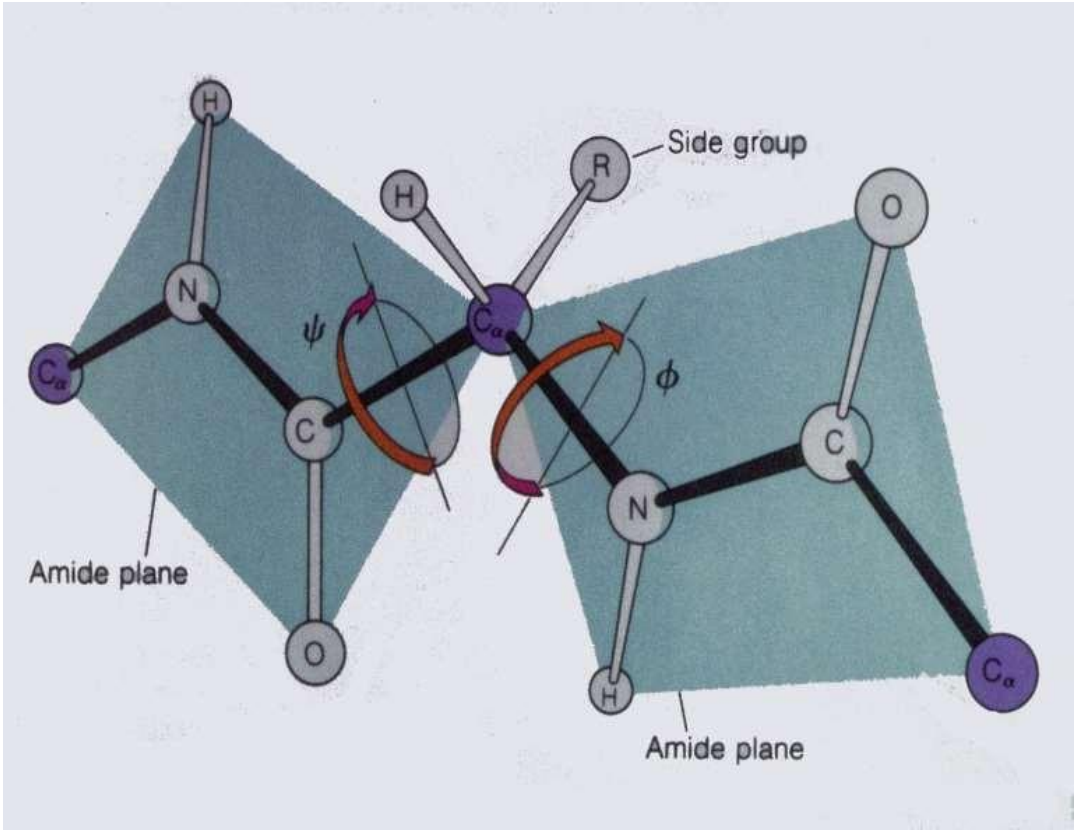




Polypeptide chain conformations

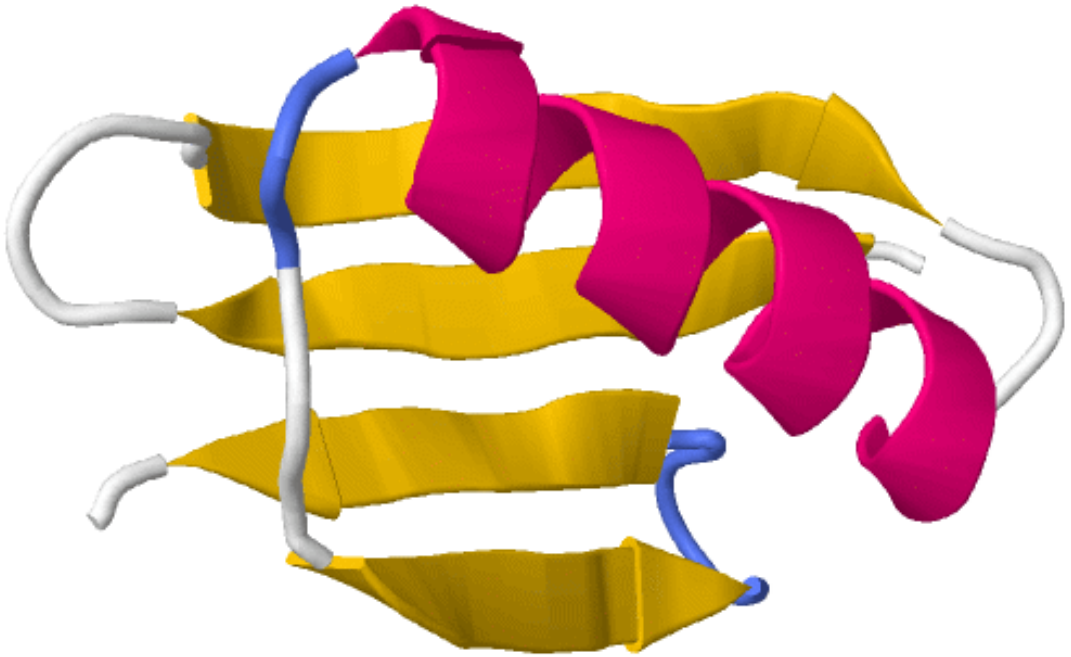
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The **Ramachandran Principle** is that **alpha helices**, **beta strands**, and **turns** are the most likely conformations for a polypeptide chain to adopt, because most other conformations are impossible due to steric **collisions between atoms**.

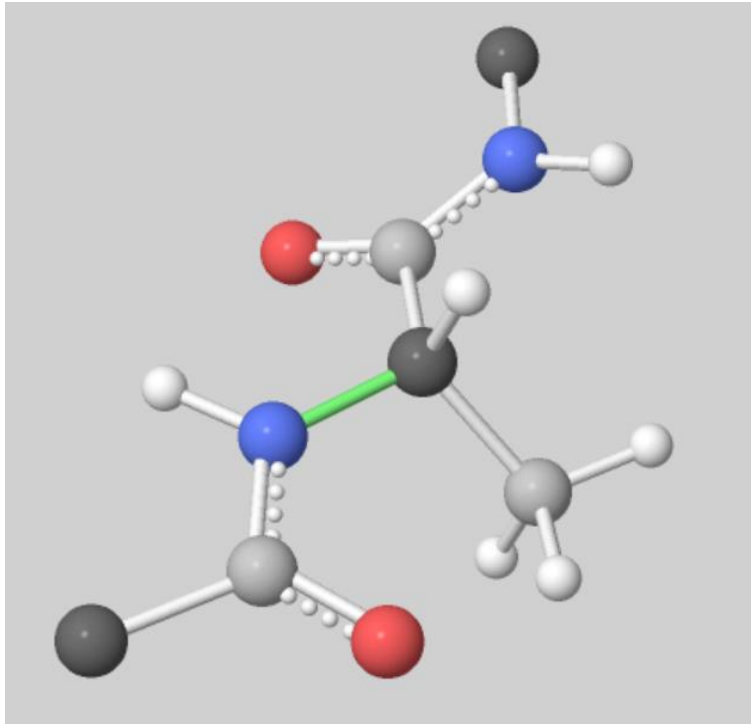
Phi (ϕ) and **Psi** (ψ) dihedral angles determine the conformation of a polypeptide.

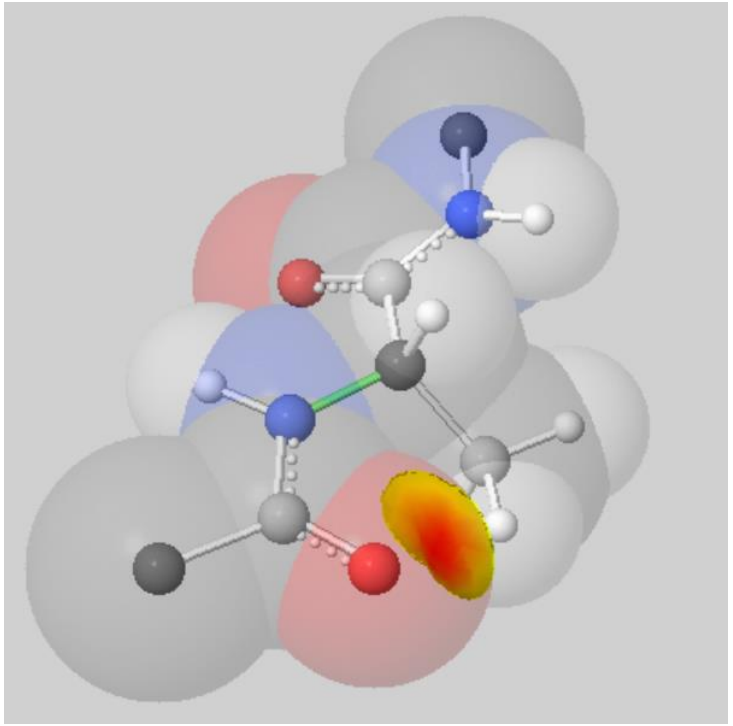


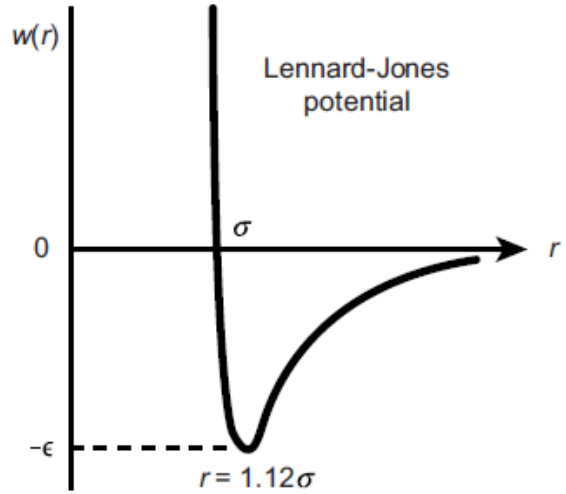
Jmol

Courtsey: Eric Martz

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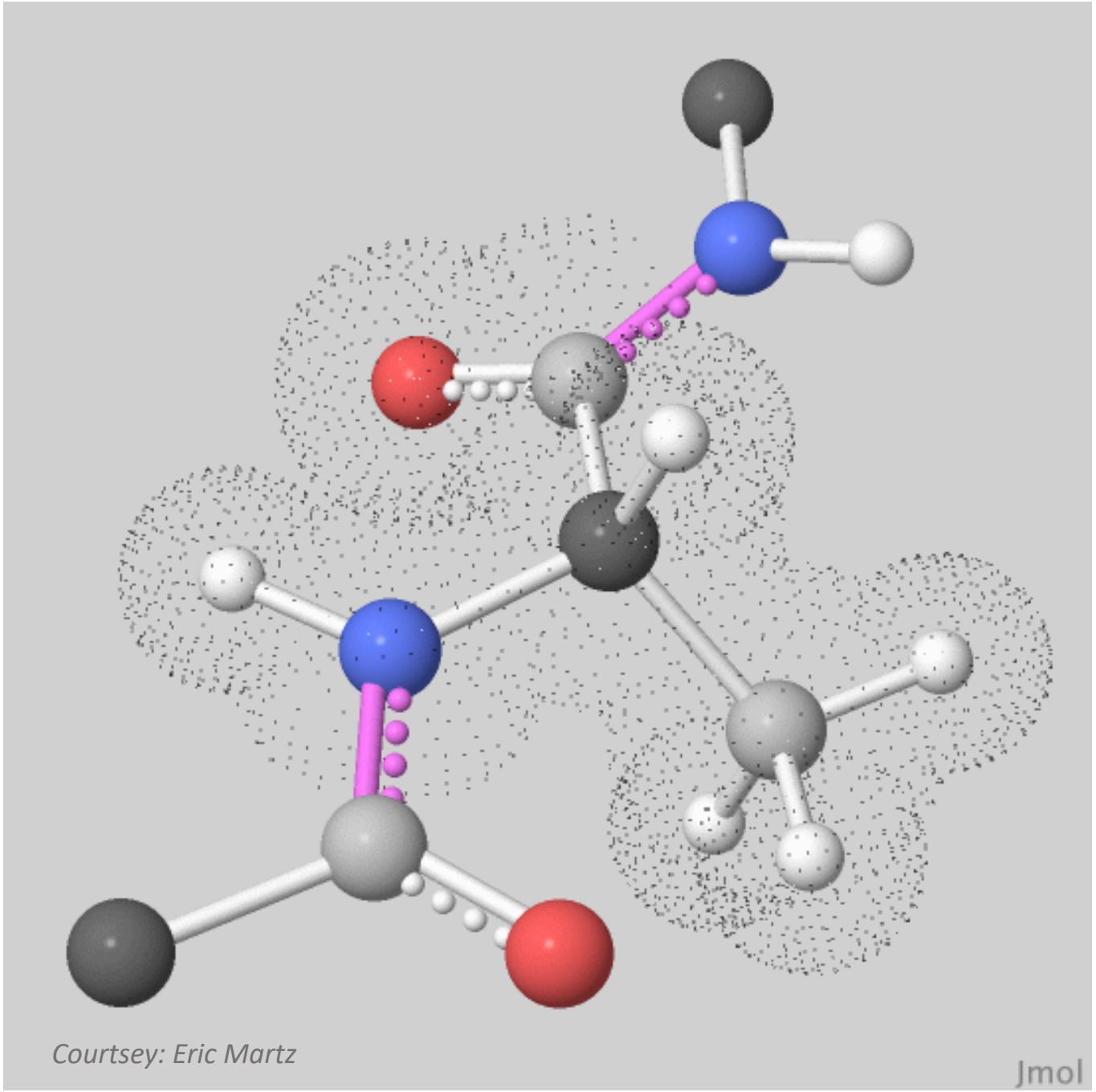






$$V(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

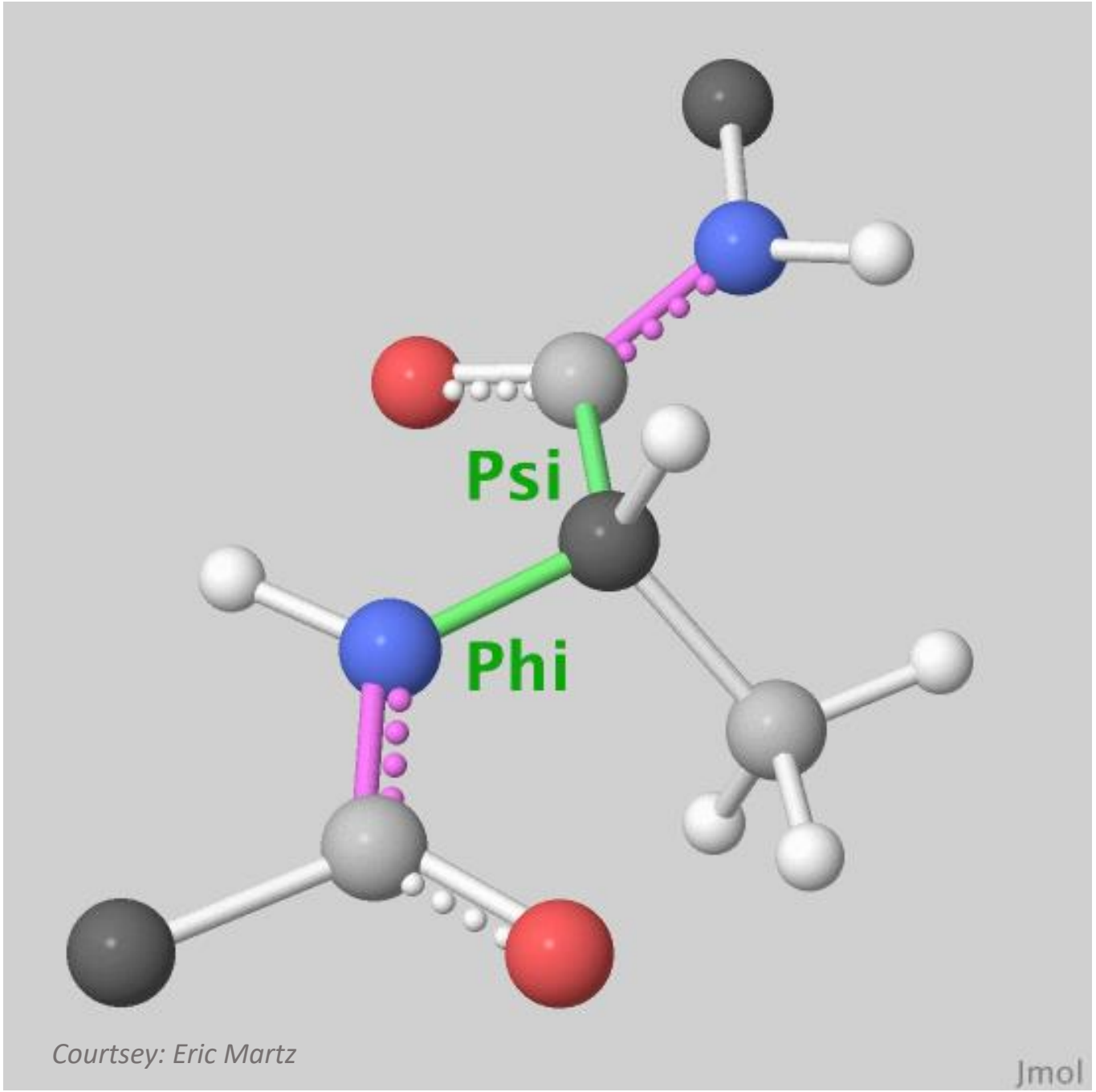
Courtsey: Eric Martz



Cα C H N O

Black dots mark a single **alanine** amino acid. Other atoms are fragments of adjacent residues in a polypeptide chain.

There are two **peptide bonds**.



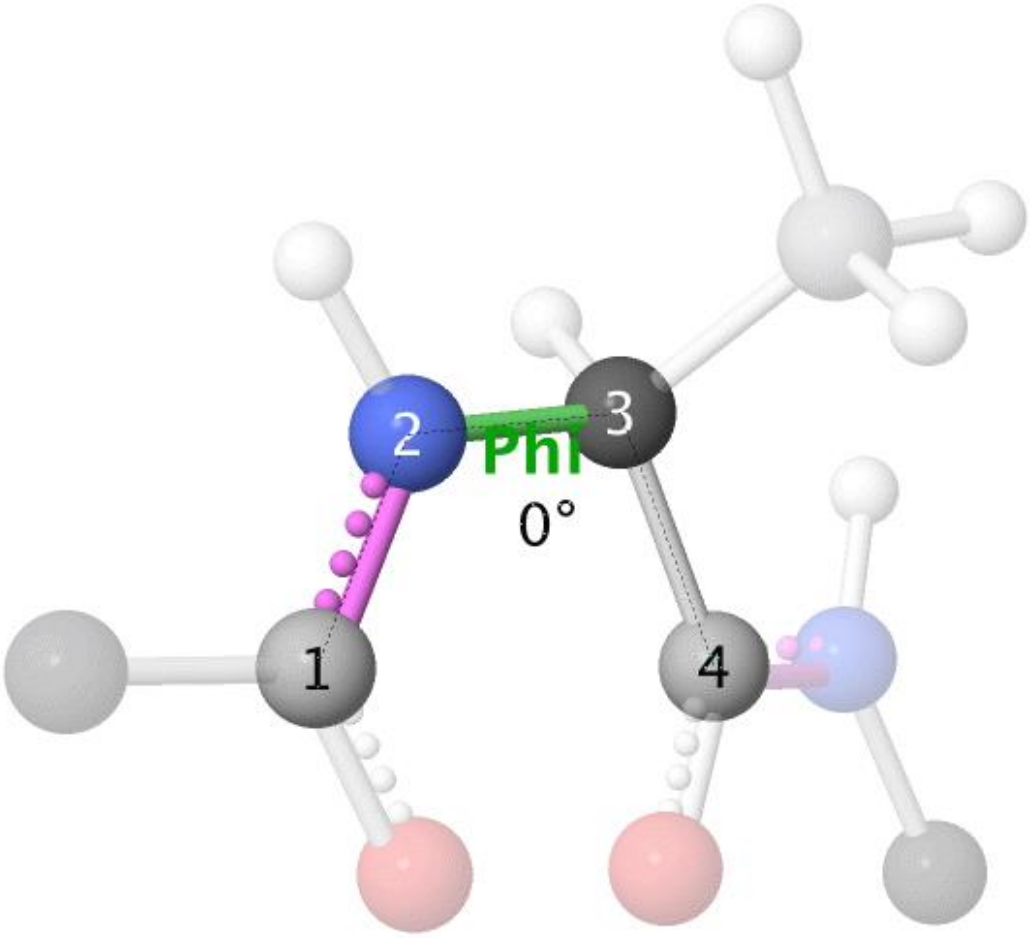
Cα C H N O

Each amino acid has two main-chain (backbone) bonds that can rotate, defining dihedral angles

Phi (ϕ) and **Psi** (ψ).

Peptide bonds cannot rotate because they are partial double bonds.

Phi = 0°



Cα C H N O

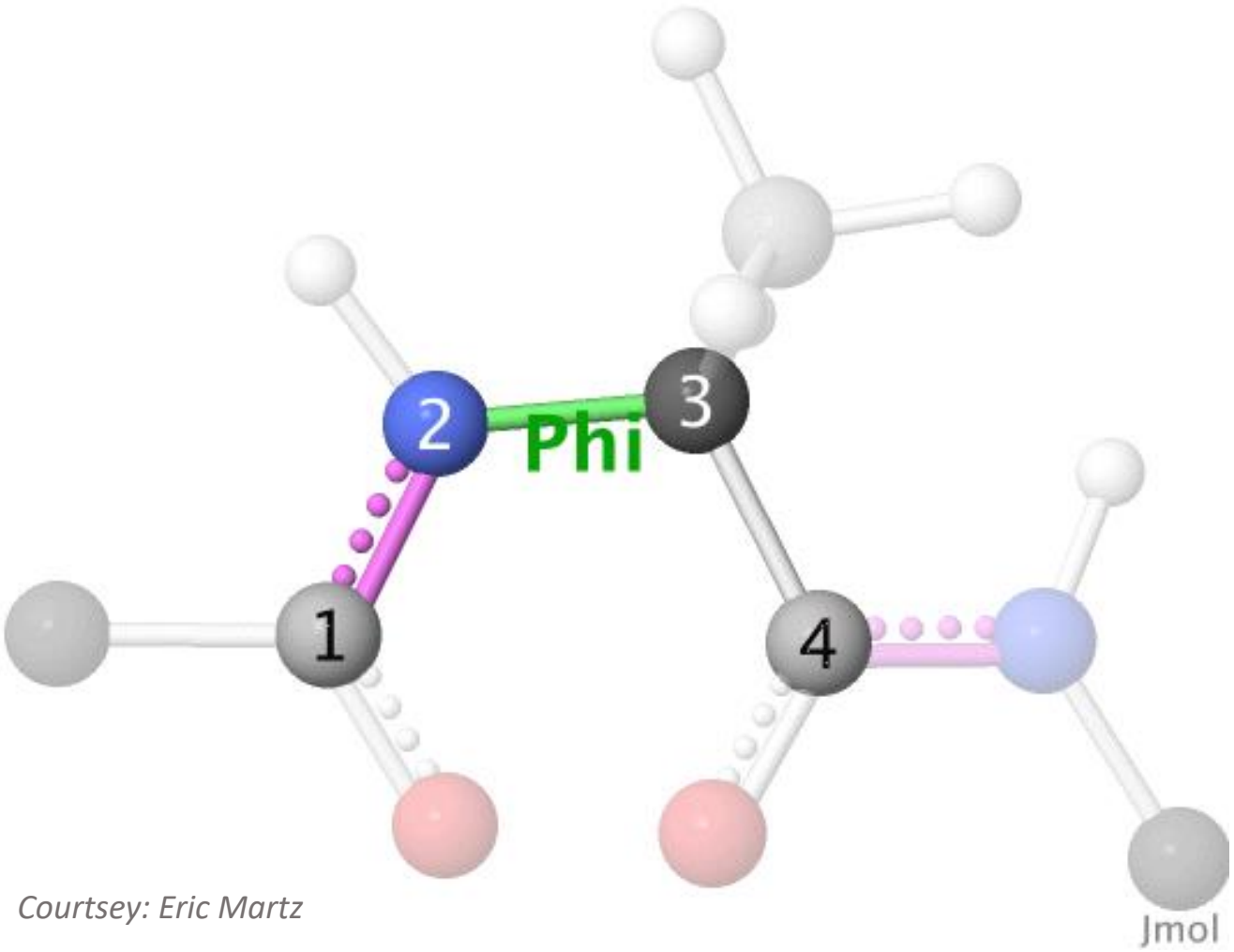
A **dihedral angle** is determined by 4 atoms. These 4 determine **Phi (ϕ).**

Here, the angle is 0º.

Courtsey: Eric Martz

Jmol

Phi = 0°

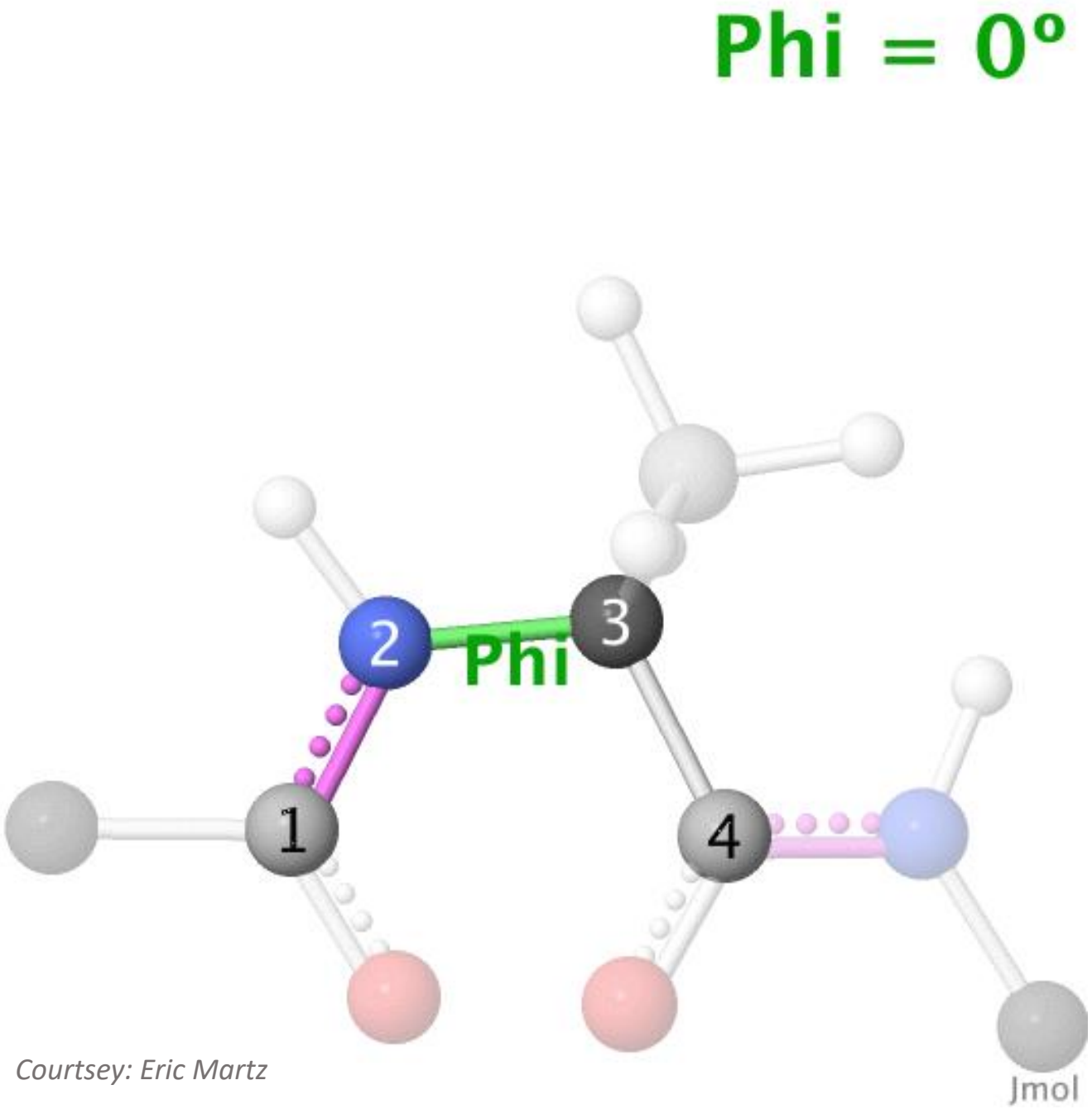


Courtsey: Eric Martz

Cα C H N O

Clockwise rotation gives dihedral angles from 0º to 180º.

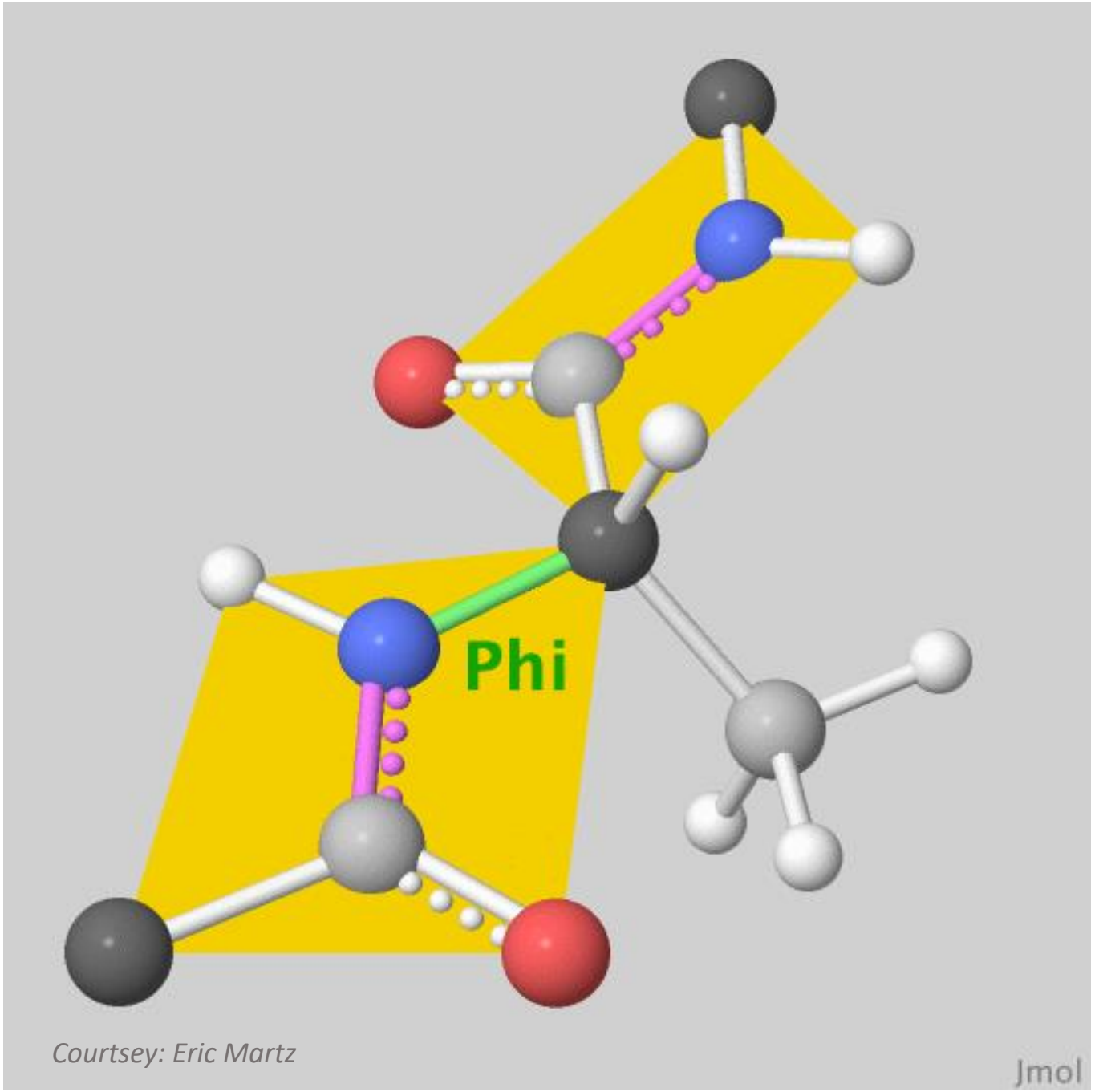
180º is the same as counter-clockwise rotation to -180º.



Courtsey: Eric Martz

Cα C H N O

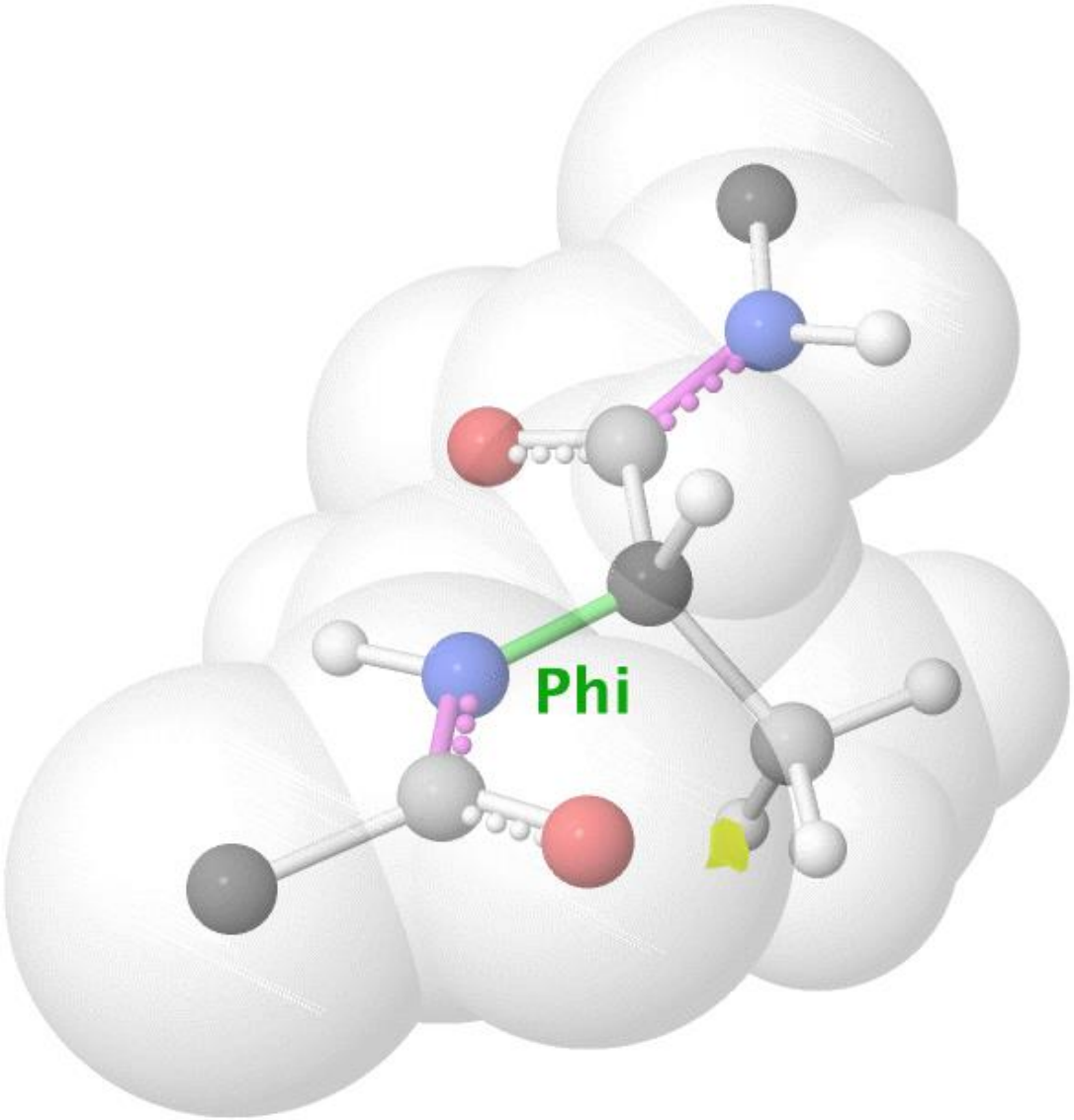
Counter-clockwise rotation starting at 0º gives *negative* dihedral angles from -1º through -90º ending at -180º (the same dihedral angle as +180º).



Cα C H N O

Each **peptide bond** holds 6 atoms in a **plane**.

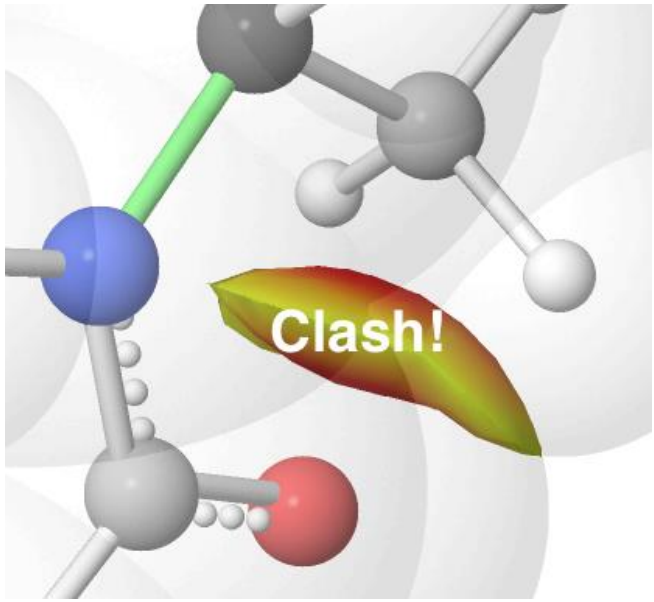
However, most possible angles of **Phi** (ϕ) and **Psi** (ψ) are impossible due to clashes between atoms.



Cα C H N O

Courtsey: Eric Martz

When **Psi** (ψ) is fixed at 165º, most **Phi** (ϕ) angles produce **clashes**:

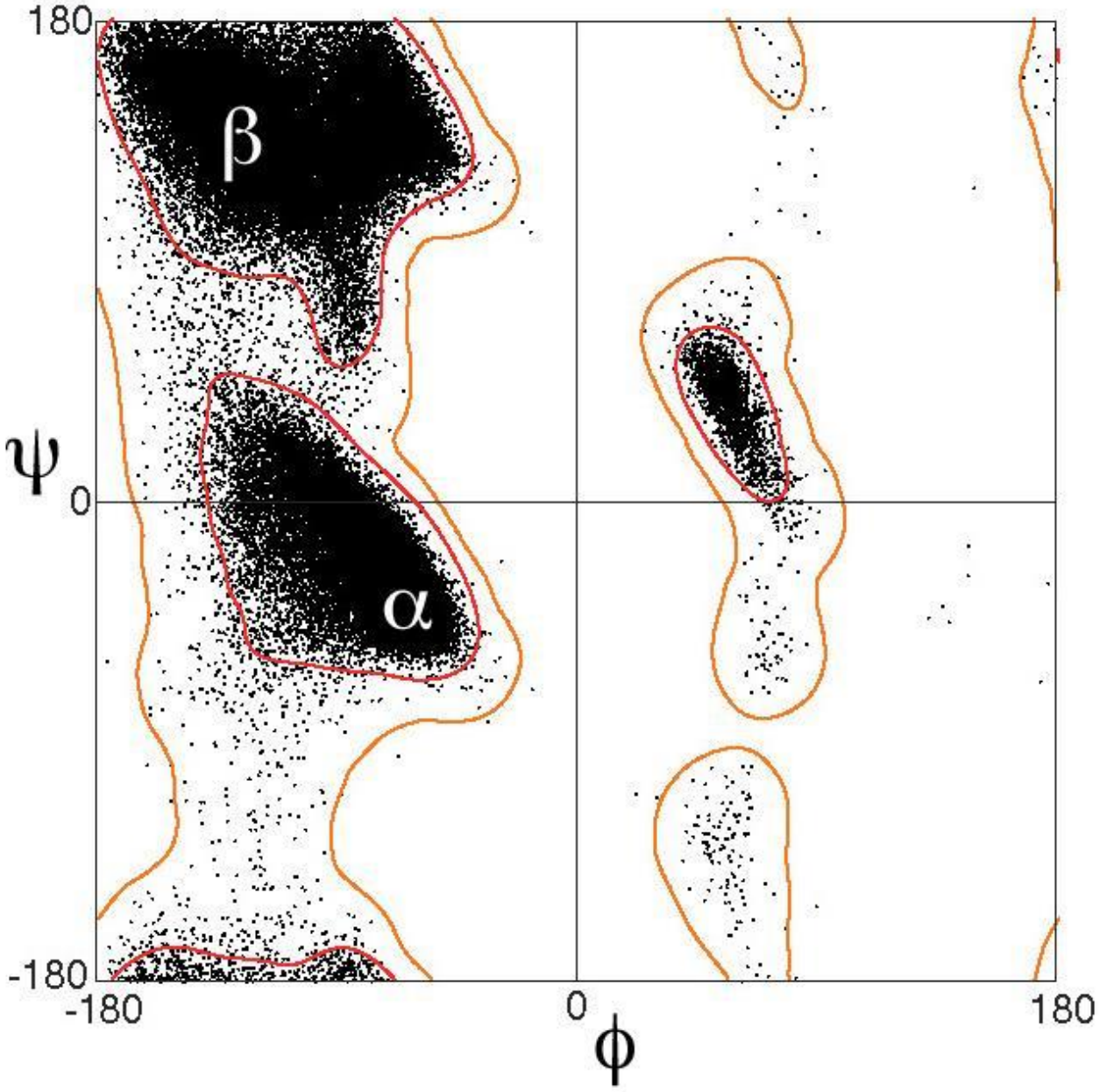


This simulation displays **clashes** while permitting van der Waals overlaps that would be physically impossible.

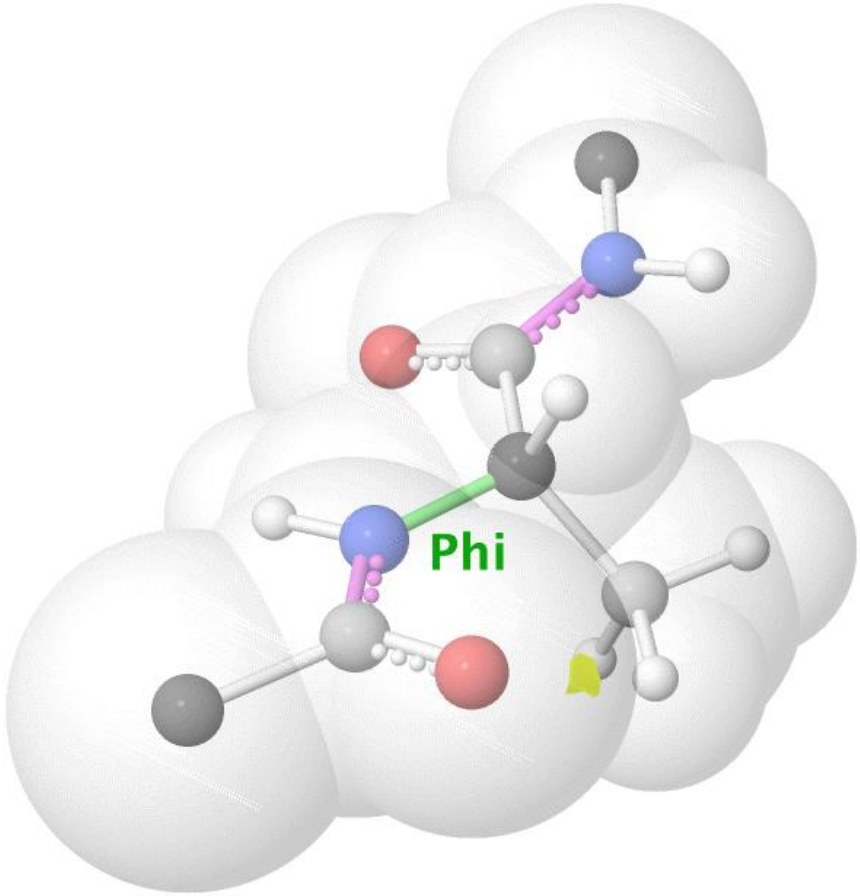
Here is a **Ramachandran Plot**. Each dot shows a pair of Phi (ϕ) Psi (ψ) dihedral angles that occurs in a protein structure. Dots cluster into alpha helical (α), beta strand (β), and turn (at right) conformations.

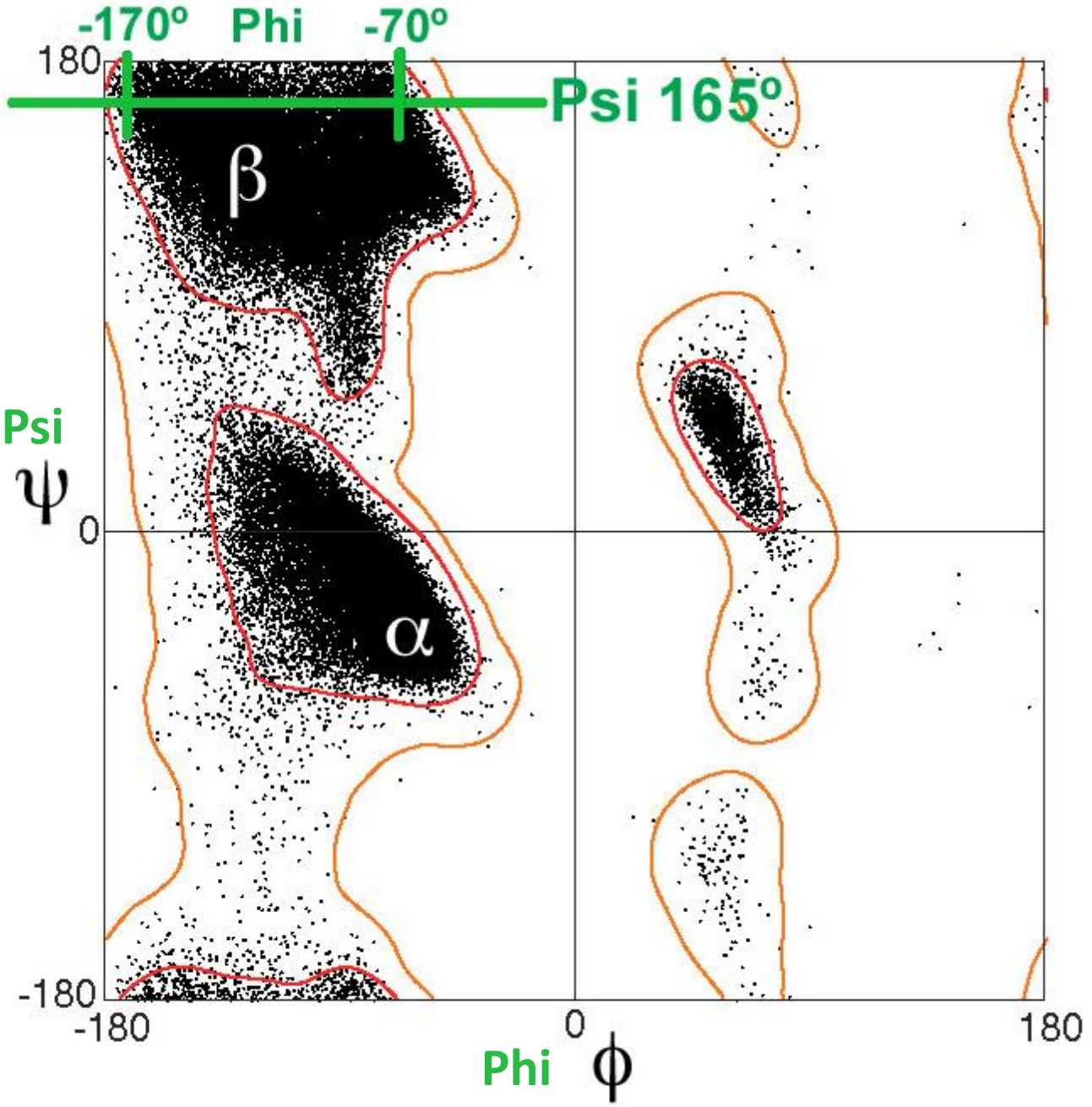
Conformations with no dots do not occur.

100,000 dots taken from high-resolution crystallographic structures. Figure from Wikimedia Commons courtesy Jane and David Richardson (Proteins 50:437, 2003). This plot excludes Gly, Pro, and pre-Pro.



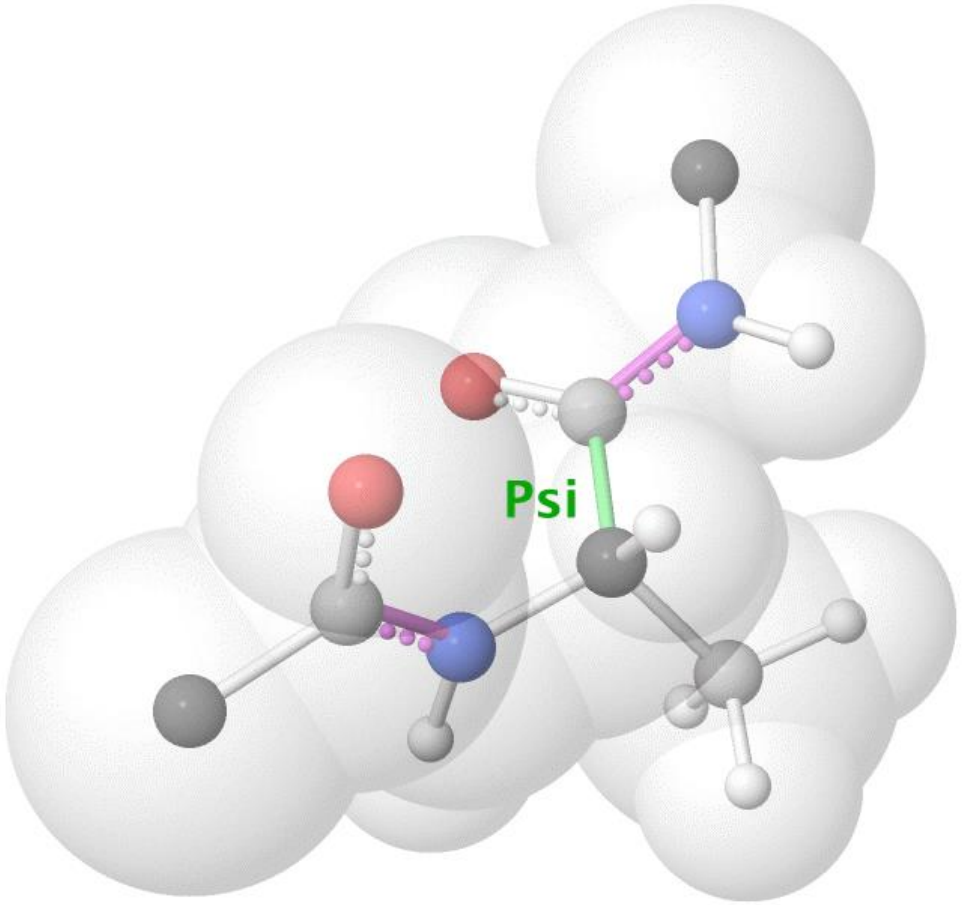
Courtesy: Eric Martz

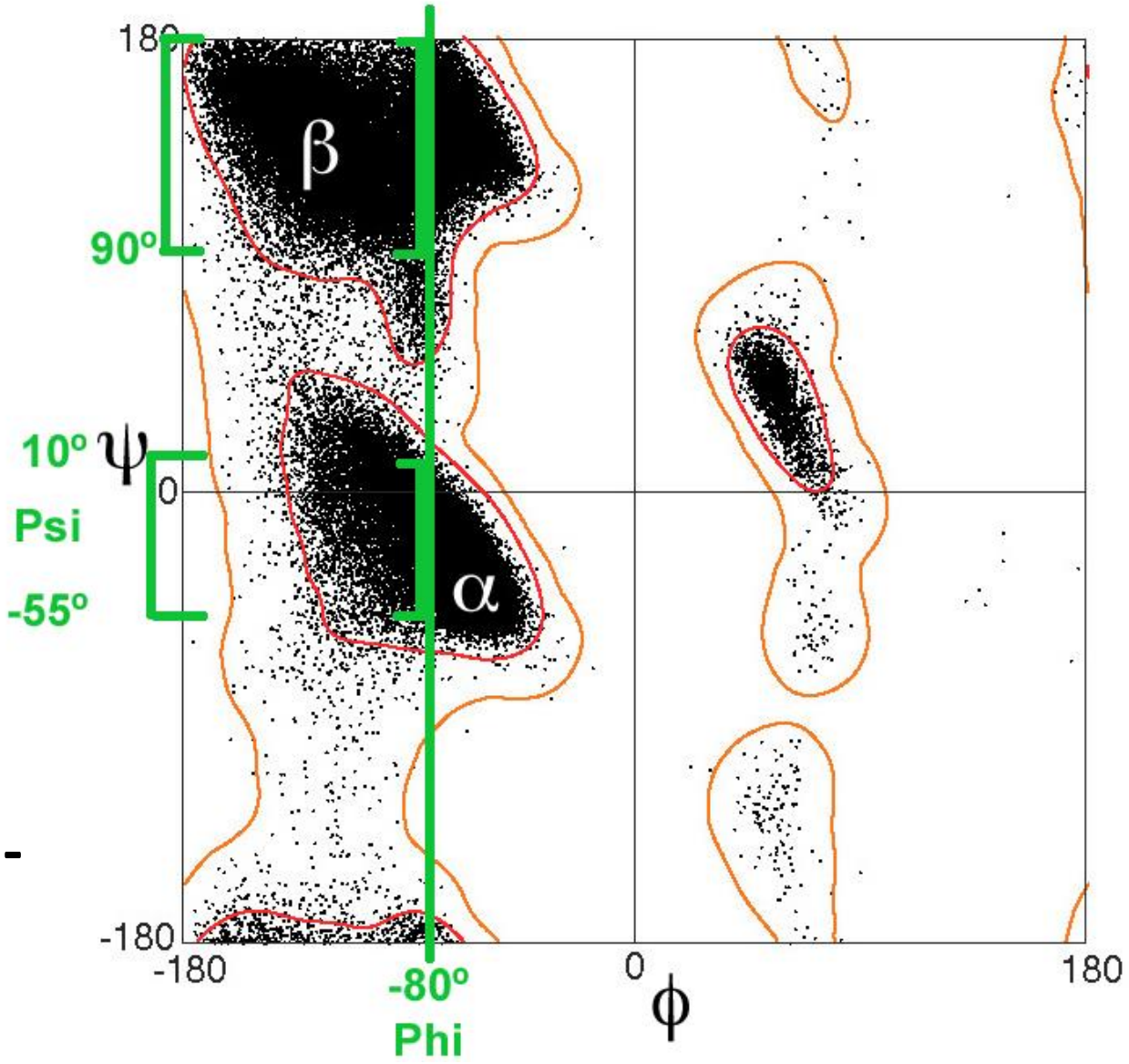




In the above animation, **Psi** is fixed at **+165º**. **Clashes** occur at all **Phi** angles except -70º to -170º. This range is in the beta strand region.

Courtsey: Eric Martz



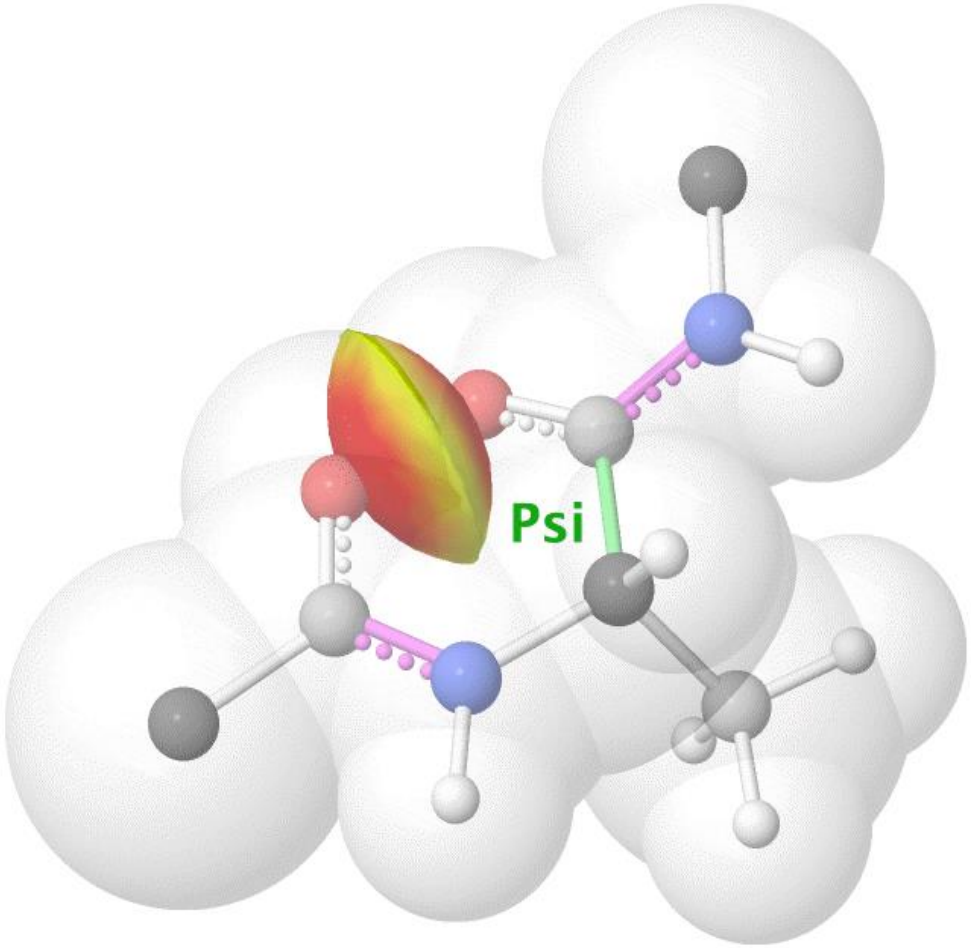


In the above animation, **Phi** is fixed at **-80º**. **Clashes** occur at all **Psi** angles except:

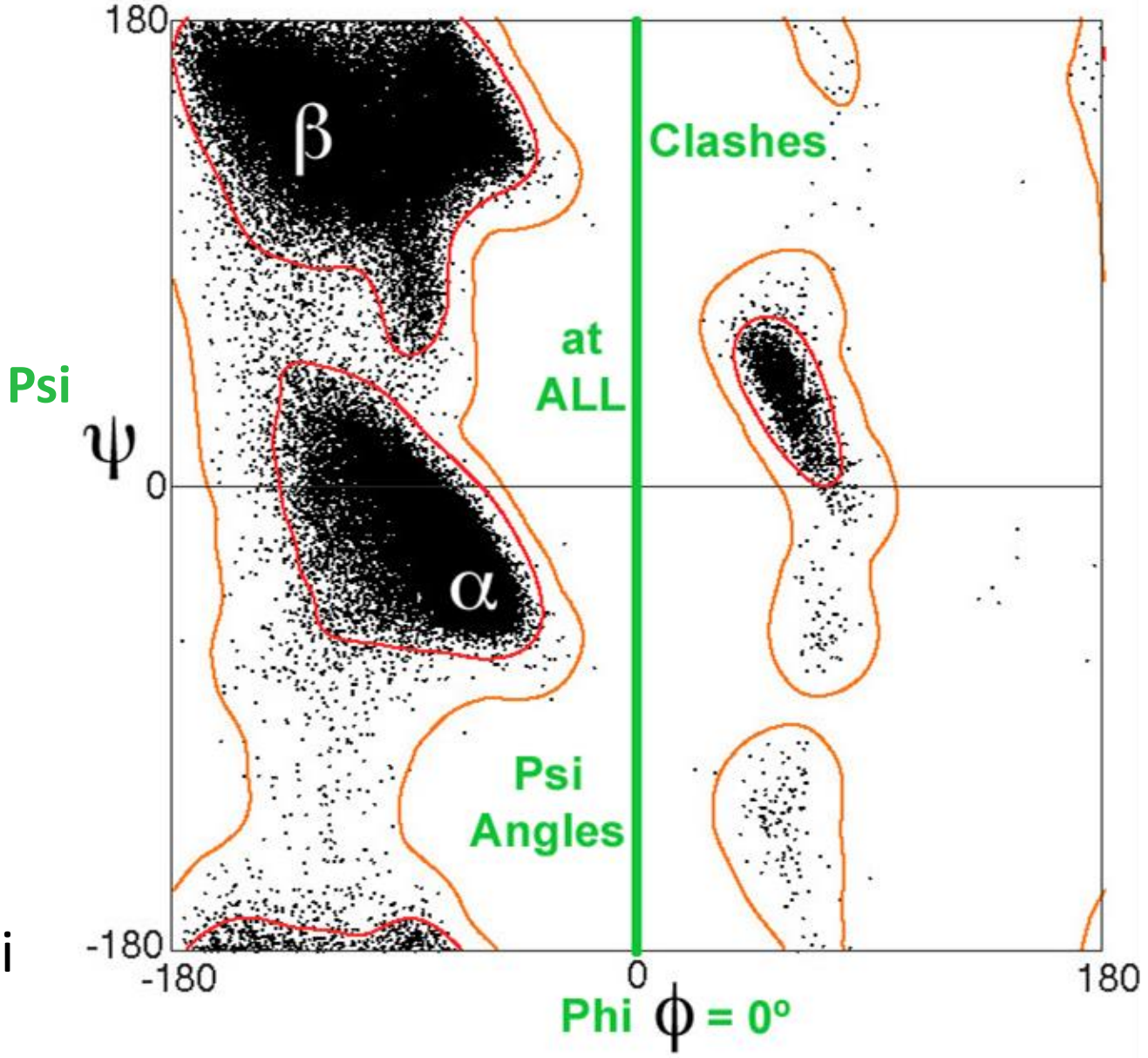
+10 to -55º (alpha helix)

+90 to 180 (beta strand)

Courtsey: Eric Martz



In the above animation, **Phi** is fixed at **0º**. **Clashes** occur at ALL **Psi** angles. Phi = 0º is physically impossible.



Courtsey: Eric Martz



Protein Folding

Protein folding and misfolding

Protein folding → provides function
Protein misfolding → improper function (gain or loss of function), can be damaging to the cell

Misfolding can result in various diseases:

- Parkinson's disease → α-synuclein aggregation
- Alzheimer's Disease → Aβ Amyloid plaques
- Familial ALS → mutant and misfolded superoxide dismutase

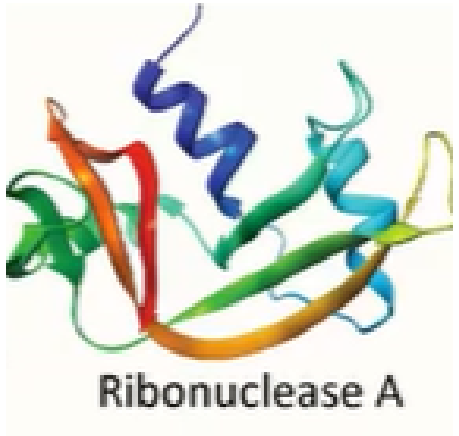
A great deal of interest is being placed on understanding protein misfolding from the point of view of disease prevention and intervention

Anfinsen's hypothesis



Christian Anfinsen – 1916-1995, Noble prize in chemistry in 1972

- Thermodynamics hypothesis of protein folding
- Seminal experiments on refolding of ribonuclease A



Anfinsen hypothesized that the primary sequence of a polypeptide dictates shape in aqueous solution – it dictates an array of possibilities and the thermodynamically most favorable result

Anfinsen's Experiment

RNase A – 124 AA long polypeptide

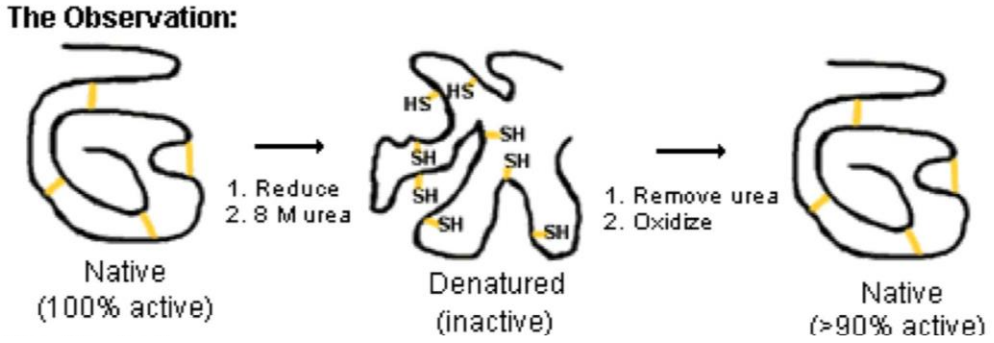
- Four disulfide bonds
- 8 Cystine residues present
- 105 different configurations possible

THE JOURNAL OF BIOLOGICAL CHEMISTRY
Vol. 237, No. 6, June 1962
Printed in U.S.A.

Side-chain Interactions Governing the Pairing of Half-cystine Residues in Ribonuclease*

"These results suggest that the native molecule is the most stable configuration, thermodynamically speaking, and that the major force in the correct pairing of sulfhydryl groups in disulfide linkage is the concerted interaction of side-chain functional groups distributed along the primary sequence. "

Anfinsen Experiment



RNase A – 124 AA long polypeptide

- Four disulfide bonds
- 8 Cystine residues present
- 105 different configurations possible

Information contained in the protein sequence is sufficient to determine protein structure!

THERMODYNAMIC HYPOTHESIS:

The native structure is the GLOBAL minimum of free energy.

Anfinsen, C.B. (1973) "Principles that govern the folding of protein chains." *Science* **181** 223-230.

Under appropriate conditions, information needed to reach the native state is encoded in the primary sequence.

What Anfinsen tells us—and what it does not

- Supports: primary sequence contains information specifying the native fold.
- Supports: the native state is associated with favorable thermodynamics under suitable conditions.
- Does not mean every protein folds unaided in a cell.
- Does not mean folding is always fast or error-free.
- Chaperones can help proteins avoid aggregation and navigate cellular conditions.

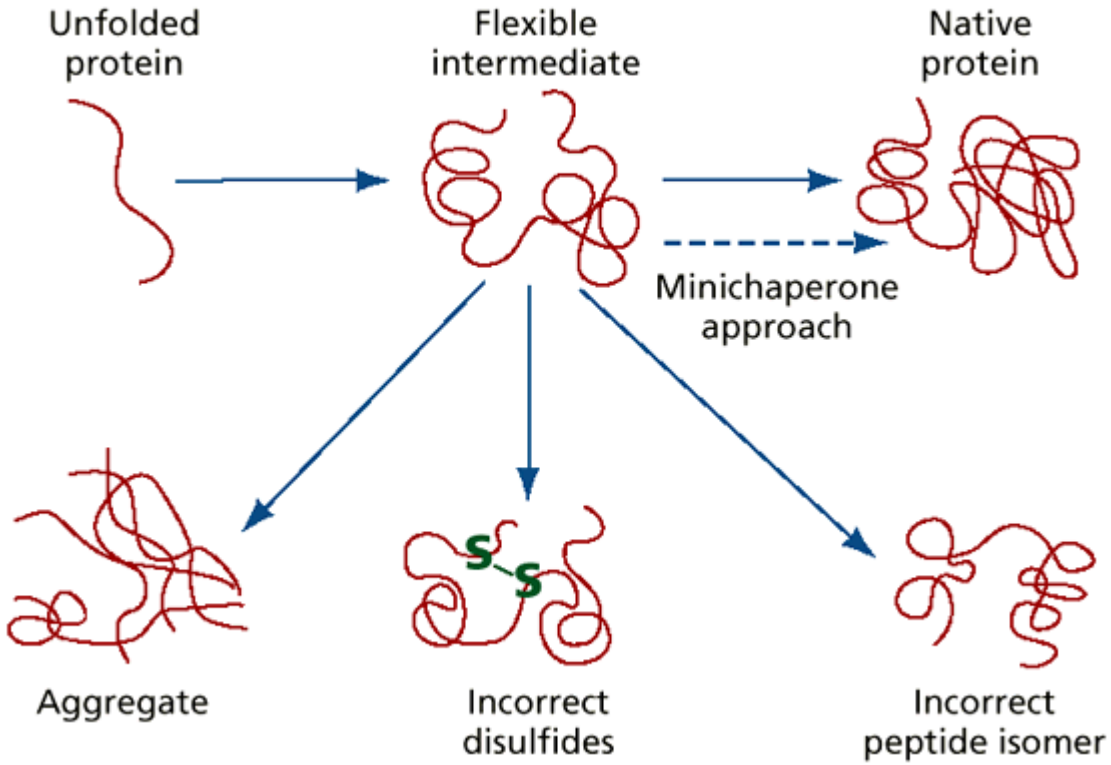
The protein-folding problem

A protein chain has many rotatable bonds.

Each bond can adopt multiple conformations.

The number of possible conformations can become astronomically large.

Yet many proteins fold rapidly and reproducibly.



https://www.nature.com/articles/nbt0299_136

Levinthal's Paradox

Protein folding – the process that results in acquisition of the native structure from a completely or partially unfolded state



Protein folding cannot proceed by purely random search among all possible conformations

For example, 100 AA protein – let's say 2 configurations for each step

It takes at minimum 10^{-12} sec for each step (time scale for bond rotation)

How long to fold?

Longer than the age of universe!

In reality…..



- Many proteins fold in seconds, sometimes need help of chaperones.
- Follow directed folding pathway
- Secondary structure fold first
- Hydrophobic collapse
  - Molten globule – much of the correct secondary structure, no tertiary structure.
- Then tertiary structure starts to fold via subomains
- Rearrangements for internal side chains via van der vals interaction.
  - Water expelled from the core
- Energy funnel is a good model.